
FULL-RANGE BINARY CLASSIFIER CALIBRATION FOR STABLE MODEL UPDATES IN PRODUCTION

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Abstract

Detection models running in adversarial environments face a malicious distribution that drifts rapidly while the benign distribution stays comparatively stable, so teams retrain and redeploy constantly to stay ahead of new threats. Retraining tends to change the output prediction scores, which breaks downstream users of the model. For these security-oriented models we need consistent false-positive rate (FPR) across all output values, whereas standard probability-calibration methods target class probability rather than an FPR contract. We introduce a method built on top of existing calibration primitives that targets the whole FPR curve, giving scores a consistent FPR meaning across deployments. On data never seen during fitting, held-out relative FPR error stays within 2.3% from 10% down to 0.1% FPR and within 7.2% at 0.01% FPR. The shipped artifact is around 200 KB and uses compiled interpolation over a 10K-knot grid, independent of calibration-set size.

1 Introduction

Detection models running in adversarial environments face a malicious distribution that drifts rapidly while the benign distribution stays comparatively stable. Teams therefore retrain and redeploy detection models continuously to stay ahead of evolving threats [1]. Retraining tends to change the output prediction scores, which breaks downstream users of the model. For these security-oriented models we need consistent false-positive performance across all output values. Standard probability-calibration methods such as Platt scaling and isotonic regression calibrate class probability rather than an FPR contract [2, 3]. We introduce a method that calibrates raw scores to the false-positive rate (FPR) directly on benign samples only, composing existing sklearn primitives (`MinMaxScaler` and `IsotonicRegression`) into a `Pipeline` that requires no custom inference code. The two-spline construction in Figure 1 keeps the FPR-to-rescaled-score lookup at fit time and ships only a raw-score-to-calibrated-score pipeline for production inference.

The FPR target depends only on the benign distribution. In the update setting we target, calibrating to FPR uses only benign traffic, which is the larger and more directly measurable sample pool, and avoids having to characterize adversarial behavior. In adversarial detection, positive examples are unknown unknowns because new attacks are hard to enumerate and label, yet their count remains far smaller than the benign count. FPR calibration is robust to positive-count uncertainty because it divides by the large, well-characterized benign count, whereas precision divides by the small, poorly characterized predicted-positive count, as quantified by the binomial planning rule in Section 2.

A single calibrated score can feed multiple downstream tiers (e.g., block at 0.1% FPR, alert at 1%, escalate at 10%), so calibration must hold across thresholds rather than only at one operating point.

The method’s contributions are:

1. **Whole-curve FPR mapping via a non-parametric monotone linear spline.** Isotonic regression over the benign empirical CDF maps score thresholds to FPR at every threshold, with no assumed functional form.

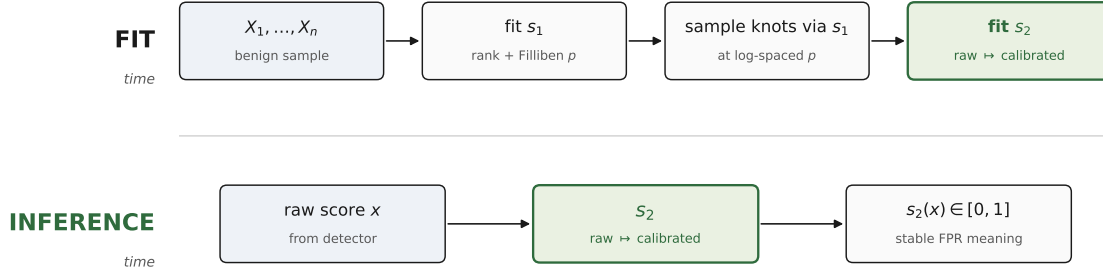


Figure 1: Method overview. During fitting, the calibrator sorts benign scores, assigns plotting-position FPR labels, and fits a temporary FPR-to-rescaled-score spline. A deterministic log-spaced FPR grid then passes through that spline and through the log-scale output contract to produce rescaled-score-to-calibrated-score storage knots. The second spline is the shipped sklearn artifact, so inference uses only raw score, fixed scaling, and the rescaled-score-to-calibrated-score map.

2. **Fixed log-scale output contract.** A piecewise-linear map in $\log_{10}(\text{FPR})$ space pins calibrated score 0.5 to 0.1% FPR, 0.7 to 0.01%, and 0.85 to 0.001%. Each step between anchors is a tenfold change in rarity, so the calibrated axis reads uniformly to an operator.
3. **Knot-subsampled, fixed-size deployable artifact.** A second isotonic fit on a deterministic log-spaced FPR grid produces a calibration artifact around 200 KB with compiled interpolation over a 10K-knot grid, independent of calibration-set size.
4. **Below-floor extrapolation with safe clipping.** Linear extrapolation from the two lowest observed (FPR, score) points, composed through the log anchors, keeps the output monotone and bounded below the sample-supported FPR floor. Calibrated score 1.0 is the terminal “flag nothing” value.
5. **Release-artifact packaging for cross-retraining stability.** The calibration artifact ships with the model, so downstream systems do not have to be recalibrated to preserve the deployment’s alert-volume budget.

Section 5 evaluates the method on the Credit Card Fraud Detection benchmark with calibration fit on a held-out subset whose benign rows are never seen during evaluation.

2 FPR estimate bounds and biases

Two finite-sample effects bound FPR estimation accuracy for any calibrator: sampling variance on the benign set and edge-of-sample bias when rank-selected score thresholds receive FPR labels for spline knots.

2.1 Sampling variance

Let S_i be the model’s scalar prediction score on benign calibration input i for $i = 1, \dots, n$, and let S_{benign} be the score the same model would assign to a fresh benign input from the same distribution. For a fixed score threshold τ , each benign calibration example either fires or does not fire, so the empirical FPR is a proportion over n independent 0/1 outcomes:

$$\hat{q}(\tau) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{S_i \geq \tau\}. \quad (1)$$

The standard normal approximation for a binomial proportion gives the 95% interval [4]

$$\hat{q}(\tau) \pm 2\sqrt{\frac{\hat{q}(\tau)(1 - \hat{q}(\tau))}{n}}. \quad (2)$$

For target FPR p and relative half-width r , converting Equation (2) gives the planning rule derived in Section A:

$$n \approx \frac{4}{r^2 p}. \quad (3)$$

Table 1 reports Equation (3) for common deployment FPRs. At the common deployment target $p = 10^{-3}$, more than 50K benign calibration samples are needed to estimate FPR within a $\pm 25\%$ relative band.

Table 1: Approximate benign sample sizes from Equation (3).

Target FPR p	$r = 50\%$	$r = 25\%$	$r = 10\%$
10^{-1}	160	640	4,000
10^{-2}	1,600	6,400	40,000
10^{-3}	16,000	64,000	400,000
10^{-4}	160,000	640,000	4.0M
10^{-5}	1.6M	6.4M	40M
10^{-6}	16M	64M	400M

2.2 Edge-of-sample bias

At a fixed score threshold, Section 2.1 measures counting error. The first calibration spline has a different problem because it starts from sorted benign scores and must attach one FPR label to each rank-selected score threshold. For a fixed threshold τ , the fresh-sample FPR is

$$q(\tau) = \Pr(S_{\text{benign}} \geq \tau), \quad (4)$$

where S_{benign} is the model score of a fresh benign deployment example.

Let $k \in \{1, \dots, n\}$ count ranks from the largest benign model score downward, and let τ_k be the k -th largest benign model score. The fresh-sample FPR at that rank-selected threshold is

$$q_k = q(\tau_k) = \Pr(S_{\text{benign}} \geq \tau_k). \quad (5)$$

The spline label attached to τ_k is written $\tilde{q}_k$ to distinguish it from the empirical FPR estimate $\hat{q}(\tau)$ in Equation (1). The naive spline label is

$$\tilde{q}_k^{\text{sample}} = \frac{k}{n}. \quad (6)$$

Equation (6) is an in-sample count: k of the n calibration scores are at or above τ_k . The spline needs the fresh-sample value q_k , the probability that a fresh benign score exceeds the threshold selected by that rank. Because τ_k was chosen after sorting, it already had to beat $n - k$ calibration scores, so a fresh benign score exceeds it less often than k/n suggests. With n sorted samples creating $n + 1$ probability gaps, the k -th largest score has mean fresh-sample FPR [5]

$$\tilde{q}_k^{\text{mean}} = \frac{k}{n + 1}. \quad (7)$$

Since $k/(n + 1) < k/n$, the naive knot assigns too large an FPR label to τ_k and returns a threshold that is slightly more selective than requested.

The mean correction shows that the naive label k/n is biased high for the fresh-sample FPR. The implementation uses Filliben because the spline must choose one typical label per rank, and Filliben is the standard median plotting-position rule. Filliben uses order-statistic medians rather than means for probability-plot locations [6], and the NIST probability-plot formula gives the interior uniform median approximation [7]. After flipping from lower-tail percentile to upper-tail FPR, the interior-rank label is

$$\tilde{q}_k^{\text{Filliben}} = \frac{k - 0.3175}{n + 0.365}. \quad (8)$$

The implementation uses Filliben by default, with endpoint values $1 - 0.5^{1/n}$ for the largest benign score and $0.5^{1/n}$ for the smallest. Held-out evaluation still reports count-based FPR; Filliben is used only to place first-spline knots during fitting.

Filliben reduces relative label bias, not relative sampling noise. At $p = 10^{-3}$ with 50,000 benign calibration samples, the relevant tail rank is about 50. Filliben moves the naive label by about 0.6% relative error, while the sampling interval from Equation (2) is about $\pm 28\%$ relative error. The correction is justified, but the practical error is dominated by having only about 50 benign tail examples. Figure B.1 shows that Filliben removes the median relative label error, and Figure B.2 shows that mean absolute relative label error barely changes because rank-selected thresholds still move across calibration sets. Only more benign calibration samples materially reduce that dominant relative error, as Table 1 shows.

3 Method

The calibrator builds a shipped raw-score-to-calibrated-score pipeline from a temporary FPR-to-rescaled-score spline and a fixed FPR-to-calibrated-score contract. Figure 1 shows the fit-time objects and the shipped object. We use `IsotonicRegression` because it is sklearn’s built-in monotone piecewise-linear spline and fits directly inside an sklearn `Pipeline`. In this method, it serves mainly as a deployable spline container. The implementation will be released at <https://github.com/cisco-ai-defense/fpr-model-calibration>.

Log-scale output anchors. Log spacing matches deployment practice because operators perceive FPR changes by order of magnitude rather than by linear differences in probability. The calibrated axis therefore starts from a fixed contract in $\log_{10}(\text{FPR})$ space before any detector-specific spline is fit. Table 2 defines the full anchor map, including the convention that a calibrated score of 0.5 corresponds to 0.1% FPR. Pinning these values gives the calibrated score the same alert-rate meaning across model versions and detector categories. The calibrated score 1.0 is reserved as a “flag nothing” sentinel because false positives can still occur when the raw score reaches its maximum, $S = 1$.

Table 2: Log-scale anchor map from FPR to calibrated score.

FPR	Calibrated	Interpretation
100%	0.00	Flag everything
10%	0.10	1 in 10 benign flagged
1%	0.30	1 in 100
0.1%	0.50	1 in 1,000
0.01%	0.70	1 in 10,000
0.001%	0.85	1 in 100,000
0.0001%	0.95	1 in 1,000,000
0%	1.00	Flag nothing

Scale scores. A `MinMaxScaler` with `feature_range=(0, 0.99)` fit on the input domain $[0, 1]$ maps raw scores to $[0, 0.99]$. Using the input domain rather than the observed calibration range keeps production scores above the calibration-time maximum inside the trained range, while reserving calibrated 1.0 as the “flag nothing” terminal.

Spline 1: FPR to rescaled score. The first spline answers the fit-time lookup from target FPR p to the rescaled score threshold that should carry it. The finite benign sample supplies thresholds only at its rank labels, so most desired FPR values on the log grid have no exact observed score. This temporary spline interpolates those missing thresholds during fitting and is not shipped. Sort the n rescaled benign scores as $s_{(1)} \leq \dots \leq s_{(n)}$. For score $s_{(j)}$, let $k = n - j + 1$ be its rank from the largest score downward. Attach the Filliben FPR label $\hat{q}_k^{\text{Filliben}}$ from Equation (8) to $s_{(j)}$. Fit `IsotonicRegression(increasing=False)` on these (FPR label, rescaled score) pairs. Every observed benign score remains a knot in this temporary spline.

Synthetic knot grid. Sample $\sim 10,000$ FPR values on a deterministic log-spaced grid, with explicit grid points at every decade from 10^{-10} to 1. The grid is log-spaced because operating ranges differ by orders of magnitude, and each FPR decade needs comparable interpolation control. For each sampled FPR, query the FPR-to-rescaled-score spline to get a rescaled-score threshold and query the fixed log-scale FPR-to-calibrated-score contract to get the calibrated score. For FPRs below Spline 1’s smallest observed FPR label, replace isotonic clipping with linear extrapolation from the two lowest observed (FPR, rescaled score) knots, bounded inside $[0, 0.99]$. These generated (rescaled score, calibrated score) pairs are storage knots for the shipped spline, not new statistical evidence below the sample-size floor.

Spline 2: rescaled score to calibrated score. Fit `IsotonicRegression(increasing=True)` on the generated (rescaled score, calibrated score) pairs. Append terminal knots (0.99, 0.99) and (1.0, 1.0) so a calibrated score of 1.0 always means “flag nothing.” If tied benign scores produce repeated score knots, `IsotonicRegression` averages the tied block while preserving monotonicity. The shipped artifact is an sklearn `Pipeline` with the `MinMaxScaler` followed by the second `IsotonicRegression`, so production inference uses standard sklearn objects rather than custom spline code.

4 Fixed-size deployable artifact

Production deployment needs a constant-size artifact and a constant-cost inference path, independent of calibration-set size. An uncapped `IsotonicRegression` fit stores one knot per unique training score, so 10M benign samples produce roughly 10M knots and a tens-of-megabytes artifact, with lookup cost proportional to training-set size.

The knot-subsampling step in Section 3 caps the shipped rescaled-score-to-calibrated-score `IsotonicRegression` at $\sim 10K$ (rescaled score, calibrated score) storage knots by construction, regardless of n . The FPR grid is deterministic log-spaced, not Monte Carlo sampled. Random subsampling of points along an order-statistic-derived curve biases the resulting calibrator toward whichever quantiles the random draw happened to land on. On a heavy-tailed log axis, that bias concentrates at the low-FPR end where calibration precision matters most. A deterministic log-spaced grid places a known count of knots in each log decade and has no selection variance. A spline passes exactly through its knot points, so approximation error is bounded by knot density in the region of interest, and log-spaced sampling gives comparable resolution across deployment FPRs that span multiple orders of magnitude.

At 10K knots the two production pipeline steps serialize under `joblib.dump` to around 200 KB; we verified the bound across calibration sets from 1K to 10M benign samples. Inference uses SciPy’s linear `interp1d` path inside sklearn, which calls NumPy’s compiled `interp` over the sorted knots for this case. Dependencies are `MinMaxScaler`, `IsotonicRegression`, and `Pipeline`, with no custom inference code: `joblib.load(path)` then `pipeline.predict(scores)`.

5 Evaluation on Credit Card Fraud Detection

We validate on the Credit Card Fraud Detection benchmark, which contains 284,807 European credit-card transactions and 492 fraud cases (0.172% positive rate) [8]. We load the dataset with `sklearn.datasets.fetch_openml(name="creditcard", version=1)`, using OpenML data_id 1597. A logistic-regression detector with standardized features is trained on a stratified 30% split (85,442 rows; 148 positives). The remaining 70% is the holdout (199,365 rows; 344 positives), scored end-to-end for every number in Table 3 and every panel in Figure 2. A 30% stratified slice of the holdout (59,809 rows: 59,706 benign and 103 positives) serves as the calibration-fit subset. The pipeline evaluates on both the full holdout and the complementary held-out-from-fit subset (139,556 rows: 139,315 benign and 241 positives) that was never seen during calibration fitting. Logistic regression is chosen over a boosted-tree baseline because its linear margin resolves deep-tail FPR cleanly. The sigmoid output of a boosted-tree classifier saturates around 10^{-3} FPR on this data and truncates the ROC tail; the calibration contract holds under either detector, but only the linear detector exercises the full anchor range.

Table 3: Training and held-out FPR at calibrated score anchors. Training FPR uses the 59,706 calibration-fit benign rows; held-out FPR uses the 139,315 benign rows not used to fit calibration. Negative relative error means held-out FPR was above target.

Calibrated score	Target FPR	Raw score	Training FPR	Held-out FPR	Rel. error
0.10	10%	0.0008	9.999%	10.053%	−0.52%
0.30	1%	0.0036	0.9999%	1.023%	−2.24%
0.50	0.1%	0.0207	0.1005%	0.1005%	−0.49%
0.70	0.01%	0.9993	0.0100%	0.0093%	+7.17%

On the held-out-from-fit subset, target FPR and held-out FPR agree within 2.3% relative error from 10% down to 0.1% FPR and within 7.2% at 0.01% FPR. Table 3 reports both the calibration-fit FPR and the held-out-from-fit FPR at each anchor from the latest run, and Figure 2 shows the corresponding full-holdout behavior across the full curve. Because its benign rows were never touched by the calibration fit, the held-out-from-fit subset measures generalization without the fit subset’s near-target FPR-by-construction. Below 0.01% the 59,706-benign fit subset hits its binomial precision floor (Table 1), and the anchor at 10^{-5} FPR would require the calibration-fit subset to scale to about 1.5M benign samples for $\pm 50\%$ planning precision.

TPR at each fixed FPR measures detection quality, not calibration quality. The detector catches 83% of fraud at 0.1% FPR, so the calibrated operating points land on real detections rather than an empty curve.

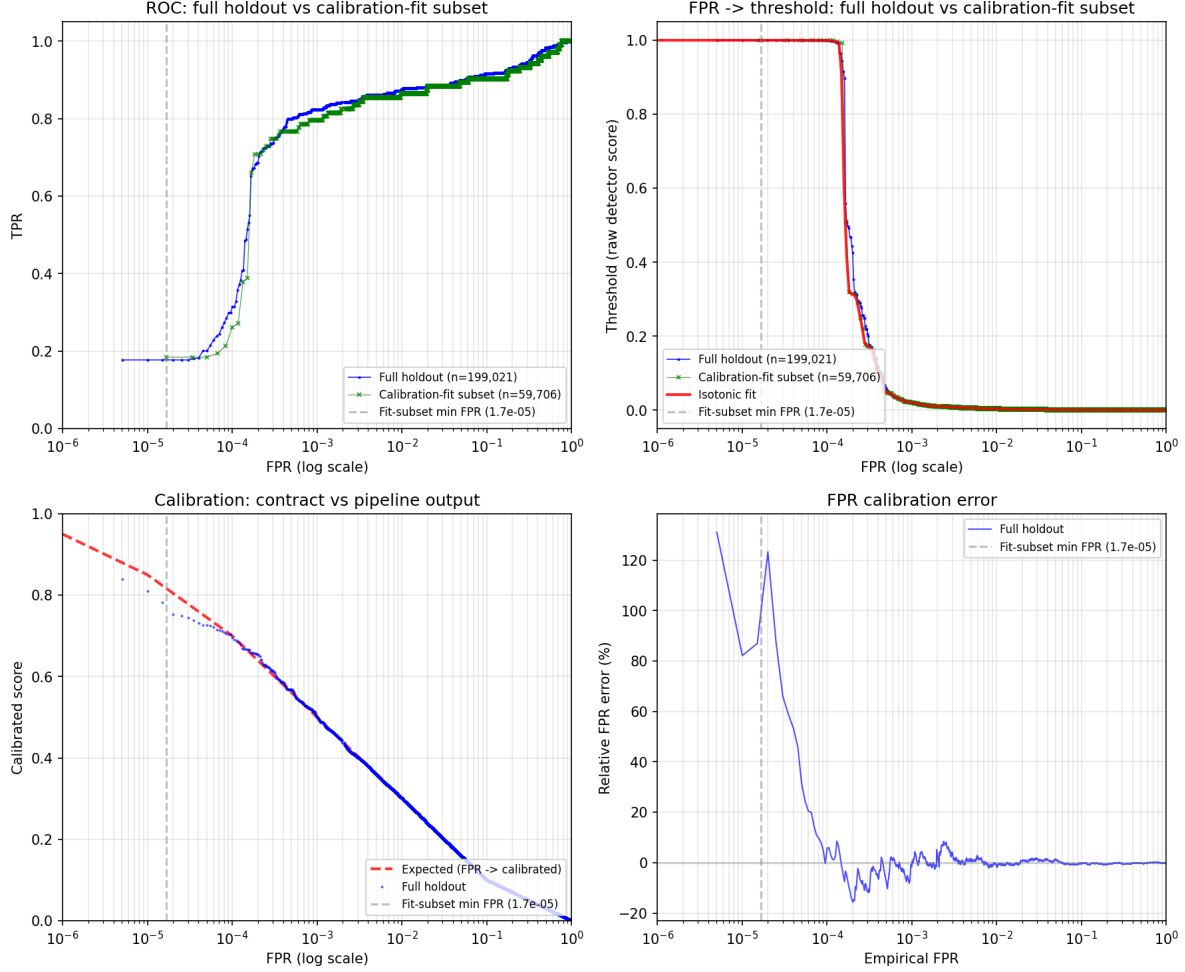


Figure 2: Four-panel validation of the calibration pipeline on the Credit Card Fraud holdout. Top-left: empirical ROC (log-FPR axis) on the full holdout in blue and on the calibration-fit subset in green (marked with $\times$), with the fit-subset’s minimum-resolvable FPR drawn as a vertical dashed line across every panel. Top-right: the empirical FPR-to-threshold relation on the same two populations, with the red line showing the fitted FPR-to-raw-score-threshold spline. Bottom-left: expected log-scale contract (red dashed) versus pipeline output on the full holdout (blue). Bottom-right: relative FPR error, computed as predicted FPR minus empirical FPR divided by empirical FPR.

A detector whose TPR collapsed at low FPR would still calibrate to the same score axis and still fail the shipping bar. That separation is deliberate: calibration keeps the FPR at each threshold consistent across deployments while detection quality is decided separately by comparing TPR at a fixed FPR across candidate models.

6 Related work

This work composes established calibration and thresholding tools into a pipeline whose released score carries an FPR meaning across the whole deployment range. Prior work supplies pieces of that construction, but not a compact score-to-FPR artifact with fixed log-FPR anchors.

Probability calibration. Platt scaling fits a sigmoid to SVM scores so they can be read as posterior class probabilities [2]. Zadrozny and Elkan use isotonic regression as a nonparametric alternative to sigmoid fitting, sorting labeled examples by score and using pool-adjacent-violators to learn a monotone step function from score to class probability [3]. Our shipped spline has the same inference shape, raw score in and calibrated value out, but the fitted quantity is the benign tail probability $q(\tau)$ from Equation (4) rather than

$\Pr(Y = 1 \mid \text{score})$. This target change matters because precision moves with the positive-class rate, while FPR is defined by benign scores only.

Outlier-score normalization. Merlion includes anomaly score calibration as a post-processing module to improve interpretability in a time-series anomaly-detection library [9]. Kriegel et al. translate arbitrary outlier factors to $[0, 1]$ values that can be compared across detectors and used in ensembles [10]. These methods output calibrated or normalized anomaly scores. Our calibrator fits the benign tail distribution directly and pins the released score to $\log_{10}(\text{FPR})$ values in Table 2.

FPR and risk control. Neyman–Pearson classifiers choose a decision rule for a specified type-I error budget. Scott and Nowak give one-sided learning-theoretic bounds of the form $R_0(\hat{h}) \leq \alpha + \epsilon$ with high probability, while Tong et al. choose an order-statistic threshold so the population type-I error exceeds α with probability at most δ [11, 12]. Learn Then Test calibrates predictive algorithms to satisfy finite-sample risk guarantees without model retraining and includes type-I outlier-error control among its examples [13]. Finn and Johnson’s early radar detector controls the threshold as a function of sampled clutter estimates, and radar textbooks treat constant-false-alarm-rate methods as threshold-detection tools [14, 15]. Inductive conformal anomaly detection computes nonconformity p-values from a proper training set and calibration scores, then raises an alarm below a chosen significance level with a false-alarm guarantee that depends on the update mode and IID assumptions [16]. Bates et al. construct conformal p-values for outlier testing and give a uniform confidence bound for FPR as the raw-score threshold varies [17]. These methods certify risk or test validity, but they do not ship a compact score transform with fixed log-FPR anchors. We estimate FPR across the score range so one released score can support block, alert, review, and logging thresholds in the same policy.

Empirical-CDF post-processing. Dadalto Câmara Gomes et al. [18] also use empirical CDFs of in-distribution detector scores, but their endpoint is a hypothesis test in which each detector score becomes a p -value under the in-distribution null and multiple detector p -values are combined with Fisher’s statistic plus Brown’s correction for correlated tests. CADET calibrates reconstruction-error anomaly scores against sample hardness and then applies a binary decision rule at an approximately tuned false-positive level [19]. Both papers calibrate anomaly or OOD scores, but their endpoint is a detector or test statistic rather than a fixed score contract for binary classifier releases.

Plotting-position corrections. Plotting positions address the rank labels used inside our first spline. The empirical label k/n is the observed calibration-set count, but a continuous benign distribution places the k -th largest score at mean fresh-sample FPR $k/(n+1)$ [5]. Filliben’s plotting positions approximate the median rank location and, after flipping from lower-tail rank to upper-tail FPR, give median-centered FPR labels [6, 7]. These corrections are upstream knot-label choices for Section 2.2, not competing calibrators.

Pipeline composition. The isotonic primitive, empirical-CDF transform, plotting-position correction, and monotone anchoring each predate this work. The pipeline combines whole-curve FPR calibration on benign scores, a fixed piecewise-linear output contract in $\log_{10}(\text{FPR})$ space, a deterministic 10K-knot artifact around 200 KB, below-floor extrapolation, and release packaging that ships the calibrator alongside the model.

7 Limitations

Score-granularity floor. A detector emitting fewer than $\sim 1/p$ distinct score values cannot be calibrated at FPR p . For target FPRs of 10^{-4} or tighter the detector itself must produce sufficiently continuous scores. On this benchmark we saw the effect directly: a boosted-tree detector saturated around 10^{-3} FPR because $\sim 0.03\%$ of benign samples shared the maximum output, preventing any lower empirical FPR from being observed regardless of threshold.

Extrapolation below the sample-size floor. Calibration below the fit set’s lowest observed tail FPR is linear extrapolation from the two lowest observed points. For production FPR targets below the floor, fit calibration on a larger benign set rather than rely on extrapolation.

Benign distribution drift. The stability of the FPR contract assumes production benign traffic resembles calibration-time benign traffic. Drift in legitimate user behavior shifts every threshold’s actual FPR. This is a data-freshness concern, not a defect of the calibrator, and it applies regardless of calibration method.

Plotting-position refinement. The implementation uses Filliben’s median plotting position [6] by default and exposes the mean-centered $k/(n+1)$ position as an option. This adjustment fixes the FPR label assigned to

rank-selected thresholds, but it does not reduce the finite-sample spread, so it does not change the sample-size requirements in Table 1.

8 Usage

Fitting, offline:

```
from fpr_model_calibration.calibration import fit_calibration_pipeline
import joblib

pipeline = fit_calibration_pipeline(benign_scores, n_knots=10000)
joblib.dump(pipeline, 'calibration.pkl')
```

Inference, production:

```
import joblib

pipeline = joblib.load('calibration.pkl')
calibrated = pipeline.predict(scores.reshape(-1, 1))
```

The log anchors ($0.5 = 0.1\%$ FPR, $0.7 = 0.01\%$, and so on) are tuned to AI security, where 1-in-1,000 is a practical planning point for automated action. For domains with a different operational block threshold, edit `_FPR_CAL_KNOTS` in `src/fpr_model_calibration/calibration.py` so that 0.5 sits at the target FPR.

9 Conclusion

FPR calibration should give operators a stable alert-rate meaning for every score threshold, not only a calibrated probability at one operating point. This method does that with benign-only data, a fixed log-scale FPR-to-calibrated-score contract, and two sklearn linear splines packaged as a standard `Pipeline`. The first spline converts target FPRs to rescaled-score thresholds during fitting. The second spline is the shipped raw-score-to-calibrated-score pipeline used at inference.

The evaluation shows that a fixed 10K-knot artifact preserves the calibrated curve when the benign calibration set supports the target FPR range, with the serialized artifact staying around 200 KB.

The main constraint is benign sample count. Filliben plotting positions place first-spline knots more accurately, but the dominant low-FPR error is still finite-sample uncertainty. The 10K-knot representation makes deployment cheap; for production targets below the observed FPR floor, the practical answer is more benign calibration data, not a more elaborate spline.

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A Sample-size planning rule

The sample-size table uses the normal approximation in Equation (2) as a planning rule. Use the Clopper–Pearson interval [20] for exact intervals. At target FPR p , the 95% half-width is approximately

$$2\sqrt{\frac{p(1-p)}{n}}. \quad (\text{A.1})$$

In the low-FPR regime, $p(1-p) \approx p$, so the half-width becomes

$$2\sqrt{\frac{p}{n}}. \quad (\text{A.2})$$

A relative half-width of r means that the absolute half-width is at most rp . Setting Equation (A.2) equal to rp and solving gives

$$2\sqrt{\frac{p}{n}} = rp \implies n = \frac{4}{r^2 p}. \quad (\text{A.3})$$

For example, $p = 10^{-3}$ and $r = 0.5$ gives $n = 16,000$ benign samples.

B Plotting-position simulation

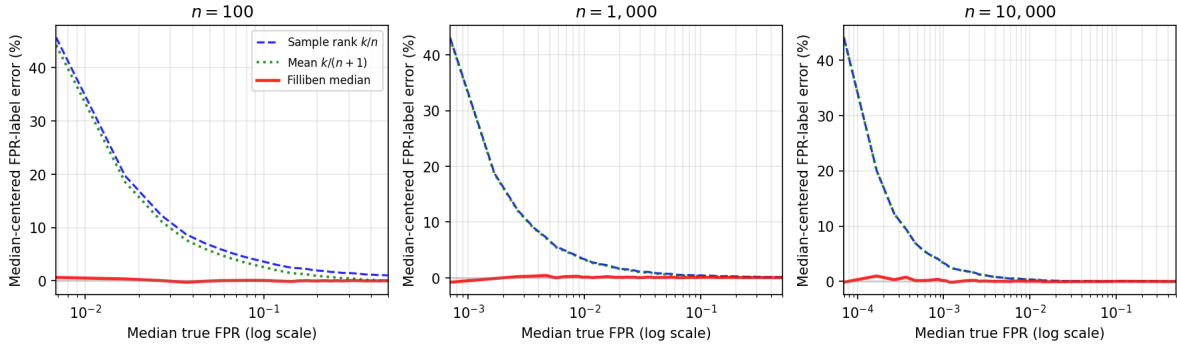


Figure B.1: Median-centered FPR-label error for threshold τ_k , the k -th largest benign model score. Each experiment draws n benign scores from $\text{Uniform}(0, 1)$, selects the threshold τ_k , and computes its fresh-sample FPR as $q_k = 1 - \tau_k$. Because the uniform distribution has survival function $\Pr(S \geq \tau) = 1 - \tau$, this measures q_k directly rather than estimating it from another finite sample. For each plotting-position rule $m \in \{\text{sample, mean, Filliben}\}$, the plotted error is $\tilde{q}_k^m - \text{median}(q_k)$, divided by $\text{median}(q_k)$, across 50,000 repeated calibration samples.

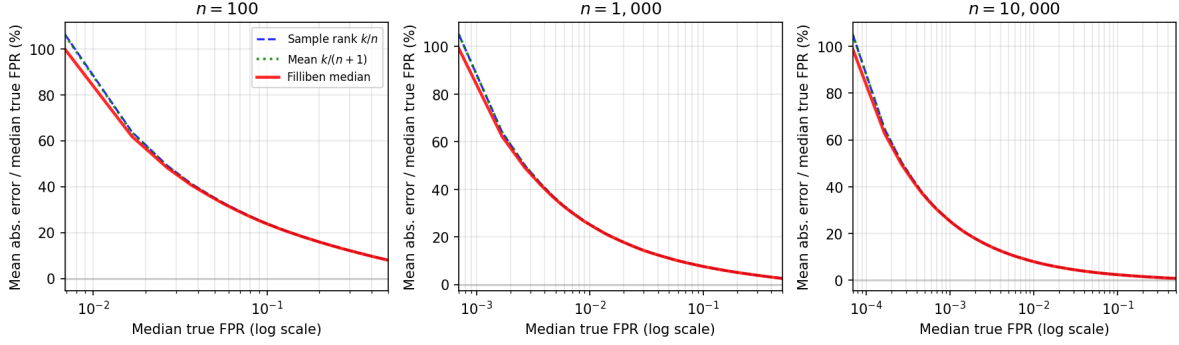


Figure B.2: Mean absolute relative FPR-label error for the same simulation as Figure B.1. For each plotting-position rule m and rank k , the plotted value is the average of $|\hat{q}_k^m - q_k|$ across repeated calibration samples, divided by the median fresh-sample FPR for that rank. The denominator uses the median fresh-sample FPR rather than the per-draw fresh-sample FPR because rare maximum-score draws can make q_k arbitrarily close to zero. This plot includes both the label-centering error and the finite-sample spread of the rank-selected threshold.